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NeuroCAPs: A Python Package for Performing Co-Activation Patterns Analyses on Resting-State and Task-Based fMRI Data

Donisha Smith[1]

1 Department of Psychology, Florida International University

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Summary

Co-Activation Patterns (CAPs) is a dynamic functional connectivity technique that clusters similar spatial distributions of brain activity. To make this analytical technique more accessible to neuroimaging researchers, NeuroCAPs, an open source Python package, was developed. This package performs end-to-end CAPs analyses on preprocessed resting-state or task-based functional magnetic resonance imaging (fMRI) data, and is most optimized for data preprocessed with fMRIPrep, a robust preprocessing pipeline designed to minimize manual user input and enhance reproducibility (Esteban et al., 2019).

Background

Numerous fMRI studies employ static functional connectivity (sFC) techniques to analyze correlative activity within and between brain regions. However, these approaches operate under the assumption that functional connectivity patterns, which change within seconds (Jiang et al., 2022), remain stationary throughout the entire data acquisition period (Hutchison et al., 2013).

Unlike sFC approaches, dynamic functional connectivity (dFC) methods enable the analysis of dynamic functional states, which are characterized by consistent, replicable, and distinct periods of time-varying brain connectivity patterns (Rabany et al., 2019). Among these techniques, CAPs analysis aggregates similar spatial distributions of brain activity using clustering techniques, typically the k-means algorithm, to capture the dynamic nature of brain activity (Liu et al., 2013, 2018).

Statement of Need

The typical CAPs workflow can be programmatically time-consuming to manually orchestrate as it generally entails several steps:

1. implement spatial dimensionality reduction of timeseries data
2. perform nuisance regression and scrub high-motion volumes (excessive head motion)
3. concatenate the timeseries data from multiple subjects into a single matrix
4. apply k-means clustering to the concatenated data and select the optimal number of clusters (CAPs) using heuristics such as the elbow or silhouette methods
5. generate different visualizations to enhance the interpretability of the CAP

While other excellent CAPs toolboxes exist, they are often implemented in proprietary languages such as MATLAB (which is the case for TbCAPs (Bolton et al., 2020)), lack comprehensive end-to-end analytical pipelines for both resting-state and task-based fMRI data with temporal

dynamic metrics and visualization capabilities (such as capcalc (Frederick & Drucker, 2022)), or are comprehensive, but generalized toolboxes for evaluating and comparing different dFC methods (such as pydFC (Torabi et al., 2024)).

NeuroCAPs addresses these limitations by providing an accessible Python package specifically for performing end-to-end CAPs analyses, from post-processing of fMRI data to creation of temporal metrics for downstream statistical analyses and visualizations to facilitate interpretations. However, many of NeuroCAPs' post-processing functionalities assumes that fMRI data is organized in a Brain Imaging Data Structure (BIDS) compliant directory and is most optimized for data preprocessed with fMRIPrep (Esteban et al., 2019) or preprocessing pipelines that generate similar outputs (e.g. NiBabies (Goncalves et al., 2025)). Furthermore, NeuroCAPs only supports the k-means algorithm for clustering, which is the clustering algorithm that was originally used and is often employed when performing the CAPs analysis (Liu et al., 2013).

Modules

The core functionalities of NeuroCAPs are concentrated in three modules:

1. `neurocaps.extraction` contains the `TimeseriesExtractor` class, which:

   - collects preprocessed BOLD data from an BIDS-compliant dataset (Yarkoni et al., 2019)
   - leverages Nilearn's (contributors, n.d.) `NiftiLabelsMasker` to perform nuisance regression and spatial dimensionality reduction using deterministic parcellations (e.g., Schaefer (Schaefer et al., 2018), AAL (Tzourio-Mazoyer et al., 2002))
   - scrubs high-motion volumes using fMRIPrep-derived framewise displacement (FD) values
   - reports quality control information related to high-motion or non-steady state volumes

2. `neurocaps.analysis` contains the CAP class for performing the main analysis, as well as several standalone utility functions.

   - The `CAP` class:
     - performs k-means clustering (Pedregosa et al., 2011) to identify CAPs, supporting both single and optimized cluster selection with heuristics such as the silhouette and elbow method (Arvai, 2023)
     - computes subject-level temporal dynamics metrics (e.g., temporal fraction, transition probabilities, etc) for statistical analysis
     - converts identified CAPs back into NIfTI statistical maps for spatial interpretation
     - integrates multiple plotting libraries (Gale et al., 2021; Hunter, 2007; Inc., n.d.; Waskom, 2021) to provide a diverse range of visualization options
   - Standalone functions: Provide tools for data standardization (Harris et al., 2020), merging timeseries data across sessions or tasks, and creating group-averaged transition matrices.

3. `neurocaps.utils` contains utility functions for:

   - fetching preset parcellation approaches (i.e. 4S, HCPex (Huang et al., 2022), and Gordon (Gordon et al., 2016))
   - generating custom parcellation approaches from tabular metadata
   - customizing plots and simulating data

Workflow

The following code demonstrates basic usage of NeuroCAPs (with simulated data) to perform CAPs analysis. A version of this example using real data is available on NeuroCAPs' readthedocs.

1. Extract timeseries data

```
import numpy as np
```

```python
from neurocaps.extraction import TimeseriesExtractor
from neurocaps.utils import simulate_bids_dataset

# Set seed
np.random.seed(0)

# Generate a BIDS directory with fMRIPrep derivatives
bids_root = simulate_bids_dataset(n_subs=3, n_runs=1, n_volumes=100, task_name="rest")

# Using Schaefer, one of the default parcellation approaches
parcel_approach = {"Schaefer": {"n_rois": 100, "yeo_networks": 7}}

# List of fMRIPrep-derived confounds for nuisance regression
acompcor_names = [f"a_comp_cor_0{i}" for i in range(5)]
confound_names = ["cosine*", "trans*", "rot*", *acompcor_names]

# Initialize extractor with signal cleaning parameters
extractor = TimeseriesExtractor(
    space="MNI152NLin2009cAsym",
    parcel_approach=parcel_approach,
    confound_names=confound_names,
    standardize=False,
    # Run discarded if more than 30% of volumes exceed FD threshold
    fd_threshold={"threshold": 0.90, "outlier_percentage": 0.30},
)

# Extract preprocessed BOLD data
extractor.get_bold(bids_dir=bids_root, task="rest", tr=2, n_cores=1, verbose=False)

# Check QC information
qc_df = extractor.report_qc()
print(qc_df)
```

	Subject_ID	Run	Mean_FD	Std_FD	Frames_Scrubbed	Frames_Interpolated	Mean_High_Motion_Length	Std_High_Motion_Length	N_Dummy_Scans
0	0	run-0	0.516349	0.289657	9	0	1.125000	0.330719	NaN
1	1	run-0	0.526343	0.297550	17	0	1.133333	0.339935	NaN
2	2	run-0	0.518041	0.273964	8	0	1.000000	0.000000	NaN

Figure 1: Quality Control Dataframe.

2. Use k-means clustering to identify the optimal number of CAPs from the data using a heuristic

```python
from neurocaps.analysis import CAP
from neurocaps.utils import PlotDefaults

# Initialize CAP class
cap_analysis = CAP(parcel_approach=extractor.parcel_approach, groups=None)

plot_kwargs = {**PlotDefaults.get_caps(), "figsize": (4, 3), "step": 2}

# Find optimal CAPs (2-20) using silhouette method; results are stored
cap_analysis.get_caps(
    subject_timeseries=extractor.subject_timeseries,
    n_clusters=range(2, 21),
    standardize=True,
```

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```
    cluster_selection_method="silhouette",
    max_iter=500,
    n_init=10,
    show_figs=True,
    **plot_kwargs,
)
```

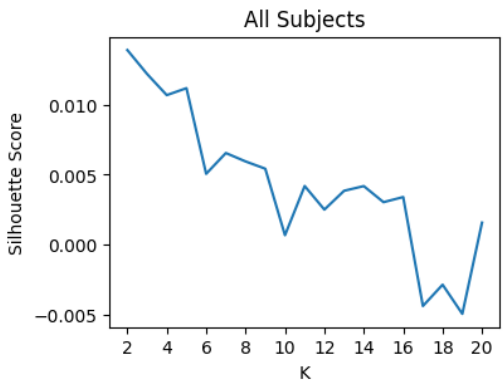

Figure 2: Silhouette Score Plot.

3. Compute temporal dynamic metrics for downstream statistical analyses

```
# Calculate temporal fraction of each CAP
metric_dict = cap_analysis.calculate_metrics(
    extractor.subject_timeseries, metrics=["temporal_fraction"]
)
print(metric_dict["temporal_fraction"])
```

	Subject_ID	Group	Run	CAP-1	CAP-2
0	0	All Subjects	run-0	0.505495	0.494505
1	1	All Subjects	run-0	0.530120	0.469880
2	2	All Subjects	run-0	0.521739	0.478261

Figure 3: Temporal Fraction Dataframe.

Note that CAP-1 is the dominant brain state across subjects (highest frequency).

4. Visualize CAPs

```
# Create surface and radar plots for each CAP
surface_kwargs = {**PlotDefaults.caps2surf(), "layout": "row", "size": (500, 100)}

radar_kwargs = {**PlotDefaults.caps2radar(), "height": 400, "width": 485}
radar_kwargs["radialaxis"] = {"range": [0, 0.4], "tickvals": [0.1, "", "", 0.4]}
radar_kwargs["legend"] = {"yanchor": "top", "y": 0.75, "x": 1.15}

cap_analysis.caps2surf(**surface_kwargs).caps2radar(**radar_kwargs)
```

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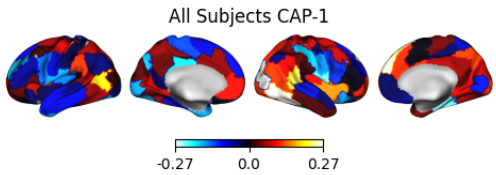

Figure 4: CAP-1 Surface Image.

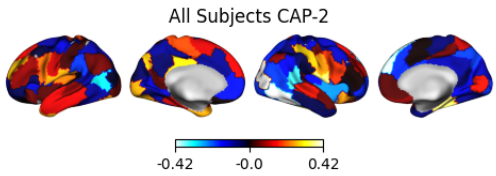

Figure 5: CAP-2 Surface Image.

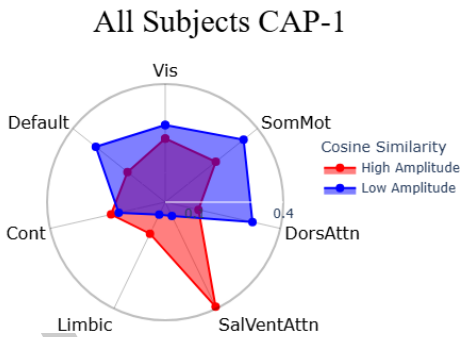

Figure 6: CAP-1 Radar Image.

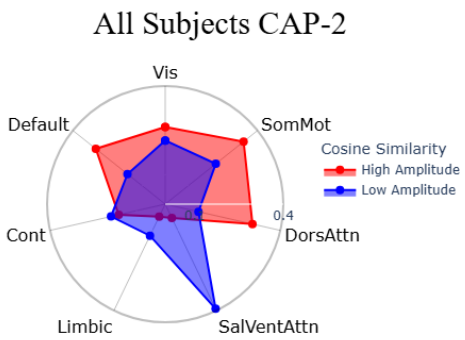

Figure 7: CAP-2 Radar Image.

Radar plots show network alignment (measured by cosine similarity): "High Amplitude" represents alignment to activations (> 0), "Low Amplitude" represents alignment to deactivations (< 0).

Each CAP can be characterized using either maximum alignment (CAP-1: Vis+/SomMot-; CAP-2: SomMot+/Vis-) or predominant alignment ("High Amplitude" – "Low Amplitude"; CAP-1: SalVentAttn+/SomMot-; CAP-2: SomMot+/SalVentAttn-).

```python
import pandas as pd

for cap_name in cap_analysis.caps["All Subjects"]:
    df = pd.DataFrame(cap_analysis.cosine_similarity["All Subjects"][cap_name])
    df["Net"] = df["High Amplitude"] - df["Low Amplitude"]
    df["Regions"] = cap_analysis.cosine_similarity["All Subjects"]["Regions"]
    print(df, "\n")
```

	High Amplitude	Low Amplitude	Net	Regions
0	0.220327	0.199130	0.021197	Vis
1	0.244341	0.154648	0.089693	SomMot
2	0.141356	0.399899	-0.258543	DorsAttn
3	0.300487	0.103134	0.197352	SalVentAttn
4	0.104964	0.092692	0.012272	Limbic
5	0.194957	0.160273	0.034684	Cont
6	0.263373	0.308228	-0.044855	Default

Figure 8: CAP-1 Network Alignment Dataframe.

	High Amplitude	Low Amplitude	Net	Regions
0	0.199130	0.220327	-0.021197	Vis
1	0.154648	0.244341	-0.089693	SomMot
2	0.399899	0.141356	0.258543	DorsAttn
3	0.103134	0.300487	-0.197352	SalVentAttn
4	0.092692	0.104964	-0.012272	Limbic
5	0.160273	0.194957	-0.034684	Cont
6	0.308228	0.263373	0.044855	Default

Figure 9: CAP-2 Network Alignment Dataframe.

Documentation

Comprehensive documentation and tutorials can be found at https://neurocaps.readthedocs.io/ and https://github.com/donishadsmith/neurocaps.

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